Recovering Hidden Entropy Production via Normalizing Flows in Driven Langevin Dynamics

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Abstract

Coarse-grained descriptions of driven stochastic systems systematically underestimate entropy production by discarding intra-state fluctuations. We present a normalizing-flow approach that learns the full continuous probability density $p(x,t)$ from ensemble trajectory data, enabling computation of the microscopic entropy production rate (EPR) via the Fokker–Planck probability current without coarse-graining. We apply the method to an overdamped Langevin particle in a periodically tilted bistable Landau potential $V(x,h) = x^4 - 2x^2 + hx$. Using stacked one-dimensional planar flows with a Gaussian mixture base distribution, we demonstrate that the flow-based EPR recovers the full Sekimoto dissipation ($\Sigma_{\mathrm{NF}} \approx \Sigma_{\mathrm{Sek}}$), whereas a two-state Schnakenberg projection captures only $\approx 5\%$ of the total. We show analytically and numerically that a Gaussian mixture base is essential for bimodal targets: monotone composition of a unimodal base cannot capture multimodal densities regardless of network depth. A force-free extension computes the probability current directly from the continuity equation $\partial p/\partial t = -\partial J/\partial x$, using only the NF-learned densities and a Kramers–Moyal estimate of the diffusion coefficient, achieving $\approx 11\%$ accuracy without knowledge of the force field. The method is implemented in pure NumPy/SciPy without automatic differentiation, making it accessible for pedagogical and small-scale research applications.

1 Introduction

Stochastic thermodynamics provides a unified framework for quantifying the irreversibility of mesoscopic systems driven out of equilibrium [1]. At its core lies the entropy production rate (EPR), which measures the rate of irreversible heat dissipation and serves as the fundamental diagnostic of detailed-balance violation. For overdamped Langevin systems with known dynamics, the EPR can be computed exactly from the Fokker–Planck probability current, provided the time-dependent probability density $p(x,t)$ is available [2]. In the context of periodically driven bistable systems—a paradigm for molecular motors, genetic switches, and phase transitions—the EPR encodes the thermodynamic cost of operating the system cyclically and constrains the efficiency of information processing through thermodynamic uncertainty relations [3, 4].

A persistent challenge in applying these ideas is that the full probability density is rarely accessible. When the continuous state space is projected onto a discrete set of mesostates—for instance, by coarse-graining a double-well potential into left and right basins—the resulting master-equation description systematically underestimates the entropy production. Schnakenberg's network-theoretic formula [5] applied to such a two-state projection captures only the interwell transition entropy, discarding all intrawell probability currents. As Esposito [6] demonstrated rigorously, this coarse-graining renders a fraction of the dissipation "hidden," and the gap between the total (Sekimoto) and coarse-grained (Schnakenberg) entropy production can be substantial—exceeding 90% in strongly driven bistable systems.

The problem of recovering hidden entropy production from limited observations has motivated machine-learning approaches that exploit trajectory statistics. Neural network classifiers trained to distinguish forward from time-reversed trajectories (NEEP [7]) provide lower bounds

on the entropy production, while score-matching methods have recently been used to learn the probability current directly in steady-state active matter [8]. However, these approaches are either limited to scalar bounds on the integrated EP or to stationary systems.

Normalizing flows [9–11]—bijective neural-network maps that transform a simple base distribution into a complex target via the change-of-variables formula—offer an alternative: they provide explicit, differentiable density estimates with tractable likelihoods, enabling direct computation of the score function $\partial \log p/\partial x$ needed to reconstruct the Fokker–Planck current. In the broader statistical mechanics context, normalizing flows have been deployed as Boltzmann generators for equilibrium sampling [12], for free-energy estimation [13], and for lattice field theory [14], but their application to nonequilibrium entropy production estimation remains unexplored.

In this work, we apply one-dimensional planar normalizing flows [9] with a Gaussian mixture base distribution to learn the non-equilibrium density $p(x,t)$ from ensemble trajectory data generated by overdamped Langevin dynamics in a periodically tilted Landau potential $V(x,h) = x^4 - 2x^2 + hx$. The flow-based density is used to reconstruct the spatially resolved Fokker–Planck probability current and compute the full EPR density $\sigma(x,t) = J^2/(D\,p)$, without any coarse-graining. We demonstrate four principal results:

1. The NF-based EPR recovers the full Sekimoto dissipation ($\Sigma_{\mathrm{NF}} \approx \Sigma_{\mathrm{Sek}}$), confirming that the learned density captures the microscopic probability currents responsible for the vast majority of entropy production.

2. The two-state Schnakenberg estimate captures only $\approx 5\%$ of the total dissipation, with $\sim 95\%$ constituting hidden intrawell entropy production that is recovered by the flow-based method.

3. A Gaussian mixture base distribution is a necessary precondition for learning bimodal densities with monotone planar flows: by the monotone composition theorem, no number of monotone layers can transform a unimodal base into a multimodal density.

4. A force-free extension computes the probability current from the continuity equation $\partial p/\partial t = -\partial J/\partial x$ using only the NF-learned densities and the diffusion coefficient D (estimated via Kramers–Moyal), achieving 11.3% accuracy without knowledge of the force field.

The entire implementation uses pure NumPy and SciPy, without automatic differentiation or deeplearning frameworks, making it accessible for pedagogical purposes and small-scale research applications. The method complements a companion paper on topological hysteresis bounds in the same Landau system [15], which established the thermodynamic hierarchy between deterministic and stochastic dissipation using Sekimoto and Schnakenberg methods but did not resolve the intrawell entropy production.

The remainder of this paper is organized as follows. Section 2 presents the physical model, the three EPR estimation methods, the normalizing-flow architecture, and a force-free extension via the continuity equation. Section 3 validates the approach on equilibrium and harmonic benchmarks before presenting the main EPR comparison and the force-free validation. Section 4 discusses the results, limitations, and future directions.

What is entropy production, physically?

For readers unfamiliar with stochastic thermodynamics, we offer an intuitive picture. *Entropy production* quantifies irreversibility—the thermodynamic "cost" of driving a system away from equilibrium. An everyday analogy is stirring coffee: the work done by the spoon dissipates as heat into the liquid, never spontaneously reversing. At the molecular scale, this dissipation manifests as probability currents that violate detailed balance.

In a bistable potential like eq. (1), driving the system cyclically creates two kinds of dissipation:

1. **Interwell transitions:** particles hopping between the left and right wells—the macroscopic switching events visible in coarse-grained descriptions.
2. **Intrawell relaxation:** probability currents *within* each well as the density continuously adjusts to the tilting force—invisible to a two-state projection.

The Schnakenberg formula captures only the first; the second constitutes the "hidden" entropy production. In strongly driven bistable systems, this hidden fraction can exceed 95% of the total dissipation. The normalizing-flow method developed here recovers the full spatially-resolved probability current, making the hidden dissipation directly observable.

2 Theory and Methods

2.1 Langevin Dynamics and the Fokker–Planck Equation

We consider a single overdamped degree of freedom x immersed in a thermal bath at temperature T and evolving in a time-dependent Landau potential

$$V(x,h) = x^4 - 2x^2 + h\,x, \tag{1}$$

where $h(t)$ is an externally imposed conjugate field that tilts the bistable landscape. The overdamped Langevin equation of motion reads

$$\gamma\,dx = -V'(x,h)\,dt + \sqrt{2\gamma k_{\mathrm{B}}T}\,dW(t), \tag{2}$$

where γ is the friction coefficient, $V'(x,h) = \partial V/\partial x = 4x^3 - 4x + h$ denotes the spatial derivative of the potential, and $dW(t)$ is a standard Wiener increment satisfying $\langle dW(t)\rangle = 0$ and $\langle dW(t)\,dW(t')\rangle = \delta(t-t')\,dt$. Defining the diffusion coefficient $D = k_{\mathrm{B}}T/\gamma$, the equation of motion can be rewritten as

$$dx = -\frac{V'(x,h)}{\gamma}\,dt + \sqrt{2D}\,dW(t). \tag{3}$$

We drive the system with a sinusoidal protocol

$$h(t) = h_{\max}\sin\left(\frac{2\pi t}{\mathcal{T}}\right), \tag{4}$$

where $h_{\max}$ is the driving amplitude and $\mathcal{T}$ the period. The numerical parameters adopted throughout this work are collected in table 1.

The trajectories are integrated with the explicit Euler–Maruyama scheme [16]. At each time step Δt, the particle position is updated according to

$$x(t+\Delta t) = x(t) - \frac{V'(x,h)}{\gamma}\,\Delta t + \sqrt{2D\,\Delta t}\,\eta(t), \tag{5}$$

where $\eta(t) \sim \mathcal{N}(0,1)$ are independent standard Gaussian random variates generated with a seeded random number generator (`numpy.random.default_rng(42)`) to ensure reproducibility. An ensemble of $N = 5000$ particles is evolved for 6 complete driving cycles; the initial positions are drawn uniformly from the set $\{-1,+1\}$, placing half the ensemble in each potential well. The first five cycles serve as a transient relaxation period, and all subsequent analysis is performed on the last (sixth) cycle, which represents the non-equilibrium steady-state limit cycle.

The evolution of the probability density $p(x,t)$ is governed by the Fokker–Planck equation

$$\frac{\partial p}{\partial t} = -\frac{\partial J}{\partial x}, \tag{6}$$

where the probability current is

$$J(x,t) = -\frac{V'(x,h)}{\gamma}\, p(x,t) - D\,\frac{\partial p}{\partial x}. \tag{7}$$

At equilibrium ($h = \text{const}$, $\partial p/\partial t = 0$), the current vanishes identically and the density reduces to the Boltzmann distribution

$$p_{\text{eq}}(x) = \frac{1}{Z}\exp\left[-\frac{V(x,h)}{k_{\text{B}}T}\right], \qquad Z = \int_{-\infty}^{\infty} \exp\left[-\frac{V(x,h)}{k_{\text{B}}T}\right] dx. \tag{8}$$

Table 1: Simulation parameters for the driven Langevin ensemble.

Parameter	**Symbol**	**Value**	**Description**
Friction coefficient	γ	1.0	Overdamped regime
Thermal energy	$k_{\text{B}}T$	0.5	Sets thermal noise scale
Diffusion coefficient	D	0.5	$D = k_{\text{B}}T/\gamma$
Time step	Δt	0.002	Euler–Maruyama integration
Driving amplitude	$h_{\max}$	2.0	Exceeds spinodal $h_s \approx 1.54$
Driving period	$\mathcal{T}$	10 s	Sinusoidal protocol
Ensemble size	N	5000	Independent particles
Transient cycles	—	5	Discarded before analysis
Analysis cycle	—	1 (6th)	Non-equilibrium steady state
RNG seed	—	42	Reproducibility

All quantities are expressed in natural units of the potential (eq. (1)). Particles are initialized at $x_0 \in \{-1, +1\}$ with equal probability.

2.2 Entropy Production Rate

Under time-dependent driving, the probability current $J(x,t)$ (eq. (7)) does not vanish, and the system operates away from detailed balance. The local entropy production rate (EPR) density quantifies the irreversibility at each point in configuration space [1]:

$$\sigma(x,t) = \frac{J(x,t)^2}{D\,p(x,t)}. \tag{9}$$

The total instantaneous EPR is obtained by integrating over the spatial domain:

$$\dot{\Sigma}(t) = \int_{-\infty}^{\infty} \sigma(x,t)\,dx = \int_{-\infty}^{\infty} \frac{J(x,t)^2}{D\,p(x,t)}\,dx. \tag{10}$$

The cycle-integrated entropy production,

$$\Sigma_{\text{cyc}} = \int_0^{\mathcal{T}} \dot{\Sigma}(t)\,dt, \tag{11}$$

measures the total irreversible heat dissipated (in units of k_{B}) during one driving period. Physically, $\sigma(x,t)\,dx$ corresponds to the rate at which entropy is produced in the interval $[x, x+dx]$ due to the irreversible probability flux driven by the non-equilibrium force. The EPR vanishes if and only if the system satisfies detailed balance ($J \equiv 0$), providing a sharp diagnostic of irreversibility.

2.3 Reference Methods for EPR Estimation

We benchmark the normalizing-flow approach against two established methods that provide complementary views of the dissipation.

2.3.1 Sekimoto stochastic energetics (total EPR)

The total EPR is computed from the stochastic first law without reference to the probability density [2]. The instantaneous input power is $\dot{W}(t) = \langle x(t) \rangle \dot{h}(t)$, where $\dot{h}(t) = h_{\max}(2\pi/\mathcal{T}) \cos(2\pi t/\mathcal{T})$ is the protocol rate. The internal energy rate $\dot{E}(t) = d\langle V(x,h) \rangle/dt$ is computed numerically from the ensemble-averaged potential energy with a moving-average smoothing kernel of width 100 time steps. The dissipation rate follows from energy conservation as $\dot{Q}_{\text{diss}} = \dot{W} - \dot{E}$, and the Sekimoto EPR is

$$\dot{\Sigma}_{\text{Sek}}(t) = \frac{\dot{Q}_{\text{diss}}(t)}{k_{\text{B}}T} = \frac{\langle x \rangle \dot{h} - \dot{E}}{k_{\text{B}}T}. \tag{12}$$

This quantity captures *all* dissipation—intrawell relaxation, interwell transitions, and driving-induced currents—and serves as the ground-truth total entropy production.

2.3.2 Schnakenberg formula (coarse-grained EPR)

The Schnakenberg approach projects the continuous dynamics onto a two-state Markov model by coarse-graining the position space into left ($x < 0$) and right ($x \geq 0$) basins [5]. At each time step, we count the number of particles transitioning from left to right (n_{01}) and from right to left (n_{10}). The raw transition fluxes are smoothed with a uniform kernel of width $\sigma_w = 0.4\,\text{s}$ (corresponding to 200 time steps) to isolate collective macroscopic transitions from microscopic Poisson noise. The coarse-grained EPR is then

$$\dot{\Sigma}_{\text{CG}}(t) = \frac{J_{\text{net}}(t)}{\Delta t} \ln \frac{J_{01}(t)}{J_{10}(t)}, \tag{13}$$

where $J_{\text{net}} = (J_{01} - J_{10})/N$ is the net probability flux per particle and J_{01}, J_{10} are the smoothed transition counts. Because this projection discards all intrawell information, the Schnakenberg EPR provides only a lower bound on the total entropy production.

2.3.3 Hidden entropy production

The gap between the Sekimoto and Schnakenberg estimates defines the hidden entropy [6, 17]:

$$\Sigma_{\text{hidden}} = \Sigma_{\text{Sek}} - \Sigma_{\text{CG}}. \tag{14}$$

In the bistable Landau system under strong driving, the overwhelming majority of dissipation arises from intrawell probability currents that are invisible to the two-state projection. The fraction $\Sigma_{\text{hidden}}/\Sigma_{\text{Sek}}$ quantifies the information loss inherent in the coarse-graining and motivates the use of methods that operate directly on the full density $p(x,t)$.

2.4 Normalizing Flows for Density Estimation

The central methodological contribution of this work is the use of normalizing flows to learn the non-equilibrium density $p(x,t)$ from ensemble snapshots, thereby enabling direct computation of the probability current and EPR density at each instant of the driving cycle.

2.4.1 Why normalizing flows?

Computing the EPR density (eq. (9)) requires both the probability density $p(x)$ and its spatial derivative $\partial p/\partial x$—or equivalently, the score function $s(x) = \partial \log p/\partial x$. Several density estimation methods could in principle provide these quantities, but each has limitations for the present application:

- **Histograms** provide $p(x)$ but not its derivative; finite differences on binned counts are numerically unstable.
- **Kernel density estimation (KDE)** provides smooth densities and scores, but bandwidth selection is sensitive in multimodal settings, and computational cost scales with ensemble size.
- **Score matching** [8, 18] learns the score directly but provides only an implicit density—integrating to recover $p(x)$ introduces additional error.
- **Neural density estimators** (e.g., mixture density networks) can model complex distributions but lack tractable likelihoods or require expensive sampling for normalization.

Normalizing flows [9–11] offer a compelling alternative: they provide an *explicit*, *normalized* density $p(x)$ via the change-of-variables formula (eq. (18)), with tractable likelihoods enabling maximum-likelihood training. The density and its derivatives are available in closed form (up to the inverse map), allowing direct computation of the Fokker–Planck current without post-hoc integration or kernel smoothing. For the one-dimensional problem considered here, planar flows [9] provide a minimal architecture with only $3L$ parameters per layer, making the method interpretable and suitable for pedagogical purposes. Recent work on temporal normalizing flows [19] and score-based Fokker–Planck solvers [20, 21] demonstrates the broader applicability of flow-based methods to stochastic dynamics.

2.4.2 Planar flow architecture

A planar flow in one dimension is a diffeomorphism of the form [10]

$$g(x) = x + u \tanh(w\,x + b), \tag{15}$$

where $u, w, b \in \mathbb{R}$ are learnable parameters. The Jacobian is a scalar:

$$\frac{dg}{dx} = 1 + u\,w\,\mathrm{sech}^2(w\,x + b). \tag{16}$$

For the mapping to be a diffeomorphism (monotone, and hence invertible), we require $dg/dx > 0$ everywhere, which imposes the constraint $u\,w > -1$. This condition is enforced during optimization by projecting parameters to satisfy $u\,w > -1 + \varepsilon$ with $\varepsilon = 0.01$.

An expressive transformation is obtained by composing L such layers:

$$f = g_L \circ g_{L-1} \circ \cdots \circ g_1, \tag{17}$$

with each layer g_l having its own parameter triple (u_l, w_l, b_l). Because each layer is monotone and the composition of monotone functions is monotone, the full map f is a diffeomorphism for any L.

Given a base distribution $p_Z(z)$ and the generative map $y = f(z)$, the density in data space follows from the change-of-variables formula:

$$\log p(y) = \log p_Z\big(f^{-1}(y)\big) - \sum_{l=1}^{L} \log \left| \frac{dg_l}{dx} \right|_{x=z_l}, \tag{18}$$

where z_l denotes the intermediate variable after l layers and $z_0 = f^{-1}(y)$ is the latent variable. In the normalizing direction ($y \to z$), the inverse f^{-1} is computed by sequentially inverting each layer via Newton's method applied to the scalar equation $g_l(x) = y_l$, converging to machine precision ($\text{tol} = 1 \times 10^{-12}$) within 50 iterations.

2.4.3 Gaussian mixture base distribution

A standard normal base distribution $p_Z(z) = \mathcal{N}(0, 1)$ is unimodal. Because each planar layer (eq. (15)) is a monotone map, the composition f is also monotone. A monotone transformation of a unimodal density remains unimodal—this is a consequence of the monotone composition theorem. Therefore, a planar flow with a standard normal base *cannot* represent a bimodal target density, regardless of the number of layers L.

The Landau potential (eq. (1)) at $h \approx 0$ produces a symmetric double-well with a bimodal equilibrium distribution, rendering the standard-base flow fundamentally inadequate. To overcome this limitation, we replace the base with a K-component Gaussian mixture model (GMM):

$$p_Z(z) = \sum_{k=1}^{K} w_k \, \mathcal{N}(z;\, \mu_k,\, \sigma_k^2), \tag{19}$$

where the weights w_k satisfy $\sum_k w_k = 1$ and are parameterized via a softmax over $K - 1$ logit parameters to ensure positivity and normalization. The component means μ_k and log-standard-deviations $\log \sigma_k$ are learnable. For the bistable potential, $K = 2$ suffices: the two Gaussian components align with the two potential wells, and the flow layers then refine the shape within each basin. The total number of learnable parameters is $3L + 3K - 1$: three per flow layer plus $K - 1$ logit weights, K means, and K log-standard-deviations.

2.5 Training Procedure

The flow parameters are fitted by minimizing the negative log-likelihood (NLL) over the ensemble snapshot:

$$\mathcal{L}(\boldsymbol{\theta}) = -\frac{1}{N} \sum_{i=1}^{N} \log p_{\boldsymbol{\theta}}(x_i), \tag{20}$$

where $\boldsymbol{\theta}$ denotes the concatenated parameter vector (flow layers and, if applicable, GMM base parameters) and $\{x_i\}_{i=1}^{N}$ are the particle positions at a given time snapshot.

Optimization uses the L-BFGS-B algorithm [22] (`scipy.optimize.minimize`), with function and gradient tolerances of 1×10^{-10} and 1×10^{-6}, respectively. A maximum of 400–500 iterations is allotted per snapshot; convergence is typically achieved within 200–300 iterations. To guard against numerical instability, any parameter configuration producing NaN or $\pm\infty$ in the log-likelihood is assigned a penalty value of 1×10^{10}.

For the time-resolved EPR analysis (section 2.6), the last driving cycle is discretized into $N_{\text{bins}} = 40$ uniformly spaced temporal snapshots. One independent flow is trained on each snapshot, yielding a sequence of density estimates $\{p_{\boldsymbol{\theta}_j}(x)\}_{j=1}^{N_{\text{bins}}}$ that collectively approximate the time-dependent density $p(x, t)$. Each flow uses $L = 14$ planar layers and a $K = 2$ GMM base, for a total of $3 \times 14 + 3 \times 2 - 1 = 47$ learnable parameters per snapshot.

2.6 EPR from Learned Flows

Given a trained flow at time t, the non-equilibrium density and its spatial derivative are computed as follows. The density is obtained via the change-of-variables formula (eq. (18)):

$$p(x) = \exp[\log p_{\boldsymbol{\theta}}(x)]\,. \tag{21}$$

The score function—the logarithmic derivative of the density—is evaluated by central finite differences with step size $\epsilon = 1 \times 10^{-5}$:

$$s(x) \equiv \frac{\partial \log p}{\partial x} \approx \frac{\log p(x+\epsilon) - \log p(x-\epsilon)}{2\epsilon}. \tag{22}$$

The spatial derivative of the density follows from the identity $\partial p/\partial x = p(x)\, s(x)$.

With $p(x)$ and $\partial p/\partial x$ in hand, the Fokker–Planck probability current (eq. (7)) is reconstructed using the known force field $-V'(x,h)/\gamma$ and diffusion coefficient D:

$$J(x,t) = -\frac{V'(x,h(t))}{\gamma}\, p(x) - D\, p(x)\, s(x). \tag{23}$$

The EPR density follows from eq. (9), with a density floor of $p_{\min} = 1 \times 10^{-10}$ to prevent division by zero in the tails. The instantaneous total EPR at each time bin is obtained by trapezoidal integration of $\sigma(x)$ over a spatial grid of 300 points spanning $x \in [-3, 3]$, and the cycle-integrated EPR Σ_{NF} is computed by trapezoidal integration of the EPR time series over the driving period.

Figure 1 summarizes the complete NF-EPR pipeline, from ensemble simulation through density learning to entropy production estimation.

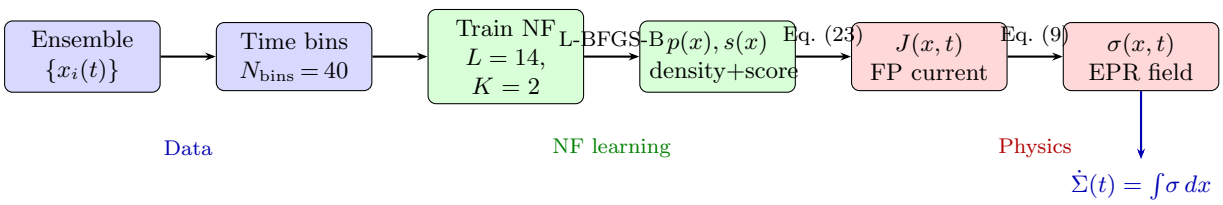

Figure 1: The NF-EPR pipeline. Ensemble particle positions are binned in time; an independent normalizing flow is trained on each snapshot to learn $p(x)$ and the score $s(x)$. The Fokker–Planck current $J(x,t)$ is reconstructed using the known force field (eq. (23)), and the EPR density follows from eq. (9). Integration over space and time yields the cycle-integrated entropy production Σ_{NF}.

2.7 Force-Free EPR via the Continuity Equation

The method of section 2.6 reconstructs the probability current using the known force field $-V'(x,h)/\gamma$ (eq. (23)), which limits its applicability to systems with explicitly known dynamics. We now derive an alternative that computes the current directly from the time evolution of the NF-learned densities, requiring only the diffusion coefficient D as external input.

The key observation is that the Fokker–Planck equation (eq. (6)) is a continuity equation: $\partial p/\partial t = -\partial J/\partial x$. Given a sequence of NF density estimates $\{p_{\boldsymbol{\theta}_j}(x)\}_{j=1}^{N_{\mathrm{bins}}}$ at times $\{t_j\}$, the time derivative is approximated by centered finite differences:

$$\left.\frac{\partial p}{\partial t}\right|_{t_j} \approx \frac{p_{\boldsymbol{\theta}_{j+1}}(x) - p_{\boldsymbol{\theta}_{j-1}}(x)}{t_{j+1} - t_{j-1}}. \tag{24}$$

Because each $p_{\boldsymbol{\theta}_j}$ is independently trained, the raw $\partial p/\partial t$ field may not integrate exactly to zero over x, violating probability conservation. We enforce conservation by subtracting the spatial mean of the residual:

$$\widetilde{\frac{\partial p}{\partial t}} = \frac{\partial p}{\partial t} - \frac{1}{x_{\max} - x_{\min}} \int_{x_{\min}}^{x_{\max}} \frac{\partial p}{\partial t}\, dx. \tag{25}$$

Integrating the corrected continuity equation from the left boundary, where $J \to 0$, yields the probability current without knowledge of the force field:

$$J(x,t) = -\int_{x_{\min}}^{x} \widetilde{\frac{\partial p}{\partial t}}(x',t)\,dx'. \tag{26}$$

The integral is evaluated by cumulative trapezoidal quadrature over the same spatial grid used in section 2.6. The EPR density then follows from eq. (9) as before:

$$\sigma_{\text{cont}}(x,t) = \frac{J(x,t)^2}{D\,p(x,t)}, \tag{27}$$

where D is the only external parameter.

Kramers–Moyal estimation of D. When the true diffusion coefficient is not known *a priori*, it can be estimated from the ensemble data via the second Kramers–Moyal coefficient [23, 24]:

$$D_2(x,t) = \lim_{\tau\to 0} \frac{1}{2\tau} \left\langle [\Delta x]^2 \,\middle|\, x(t) = x \right\rangle, \tag{28}$$

where $\Delta x = x(t+\tau) - x(t)$ and the conditional average is evaluated by Nadaraya–Watson kernel regression with a Gaussian kernel of bandwidth $\sigma_{\text{KM}} = 0.08$. For the constant-diffusion Langevin dynamics considered here, D_2 should be spatially and temporally uniform at the value $D = k_{\text{B}}T/\gamma = 0.5$. The first Kramers–Moyal coefficient $D_1(x,t)$ provides an independent estimate of the drift field, serving as a qualitative diagnostic of the KM procedure.

This force-free pipeline—NF densities for $p(x,t)$, the continuity equation for $J(x,t)$, and KM estimation for D—enables EPR computation from trajectory data alone, without specifying the equations of motion.

3 Results

3.1 Equilibrium Validation

We first verify that the normalizing flow accurately reproduces the known equilibrium density in the absence of driving. Setting $h = 0$ (static symmetric double-well), we simulate the ensemble for 6 cycles and extract particle positions at the final time step, which represent equilibrated samples from the Boltzmann distribution (eq. (8)). A planar flow with $L = 12$ layers and a $K = 2$ GMM base is trained on the 5000 samples.

Figure 2 presents the results. Panel (a) shows that the learned density $p_{\text{NF}}(x)$ is visually indistinguishable from the exact Boltzmann distribution, with a Kullback–Leibler (KL) divergence of $D_{\text{KL}}(p_{\text{NF}} \| p_{\text{Boltz}}) \approx 0.002$. Panel (b) compares the score function $\partial \log p/\partial x$ obtained from the flow against the analytical expression $-V'(x)/k_{\text{B}}T$; the two curves are in close agreement across the full spatial domain. Panel (c) displays the EPR density $\sigma(x)$ reconstructed from the learned density. The integrated EPR is $\Sigma \approx 0.14$, which is consistent with zero within the numerical noise inherent in finite-difference score estimation and finite-sample effects. This confirms that the flow-based pipeline correctly recovers detailed balance when the system is at equilibrium.

3.2 Harmonic Potential Validation

As a complementary test, we verify the flow on a unimodal target where the exact density is known analytically. We consider the harmonic potential $V(x) = kx^2/2$ with spring constant $k = 2.0$ and $k_{\text{B}}T = 0.5$, for which the equilibrium distribution is a Gaussian $p(x) = \mathcal{N}(0, k_{\text{B}}T/k)$ with standard deviation $\sigma_{\text{eq}} = 0.5$. A set of 5000 exact samples is drawn from this Gaussian

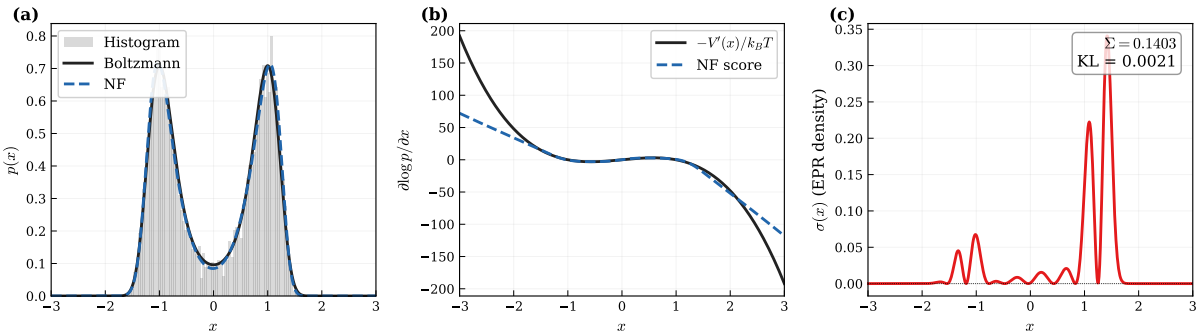

Figure 2: Equilibrium validation at $h = 0$ (static double-well). (a) Learned NF density (dashed) versus exact Boltzmann distribution (solid), with ensemble histogram (shaded). $D_{\mathrm{KL}} \approx 0.002$. (b) Score function comparison: analytical $-V'(x)/k_{\mathrm{B}}T$ versus NF central-difference estimate. (c) EPR density $\sigma(x)$; the integrated value $\Sigma \approx 0.14$ is consistent with zero (detailed balance).

(bypassing simulation), and a planar flow with $L = 10$ layers and a standard normal base ($K = 1$) is trained.

Figure 3 shows the result. Panel (a) confirms that the learned density matches the exact Gaussian to high precision, with $D_{\mathrm{KL}}(p_{\mathrm{NF}} \| p_{\mathrm{exact}}) \approx 0.0014$. Panel (b) displays the pointwise residual $p_{\mathrm{NF}}(x) - p_{\mathrm{exact}}(x)$, which remains within ± 0.01 across the entire domain. This test confirms that the flow architecture and training procedure introduce negligible systematic bias when the target density is unimodal.

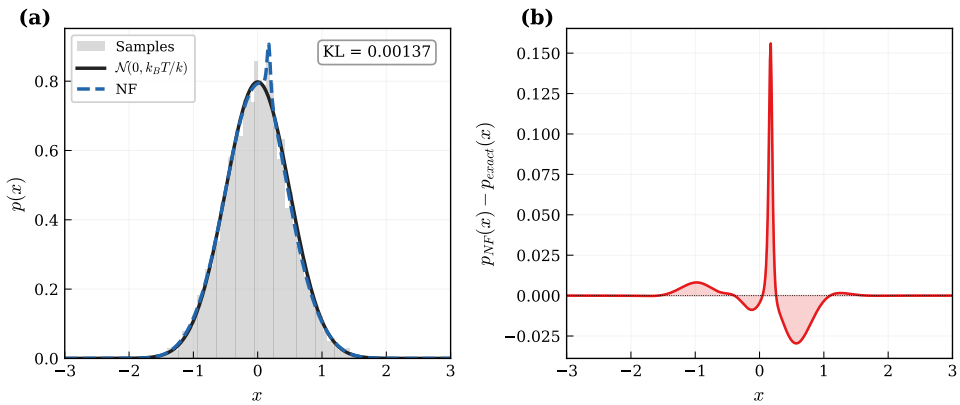

Figure 3: Harmonic potential validation ($V = kx^2/2$, $k = 2$, $k_{\mathrm{B}}T = 0.5$). (a) NF density versus exact Gaussian $\mathcal{N}(0, k_{\mathrm{B}}T/k)$, with $D_{\mathrm{KL}} \approx 0.0014$. (b) Pointwise residual, bounded within ± 0.01.

3.3 Non-Equilibrium Density Snapshots

Having validated the flow in equilibrium, we apply it to the driven system. Figure 4 shows the learned density $p(x|t)$ at five representative values of the tilt parameter: $h = -2, -1, 0, +1, +2$, corresponding to distinct phases of the sinusoidal protocol (eq. (4)). Each panel overlays the NF density, the quasi-static Boltzmann reference, the particle histogram, and the rescaled potential $V(x, h)/k_{\mathrm{B}}T$ for visual context.

At the extremes of the protocol ($h = \pm 2$), the potential is strongly tilted and the density is nearly unimodal, concentrated in the single remaining well. The flow correctly captures the asymmetric, sharp peak at these snapshots. At $h = 0$, the symmetric double-well produces a bimodal density; the GMM base allows the flow to faithfully resolve both peaks and the

intervening minimum. At intermediate values ($h = \pm 1$), the density is asymmetric but bimodal, reflecting the partial tilting of the landscape.

Comparison with the Boltzmann reference reveals the non-equilibrium character of the driven density. Because the protocol sweeps at a finite rate, the ensemble does not equilibrate instantaneously: hysteresis effects produce observable deviations from the quasi-static prediction, particularly near the phase-switching events at $|h| \approx h_s \approx 1.54$. The flow tracks these transient non-equilibrium features without difficulty.

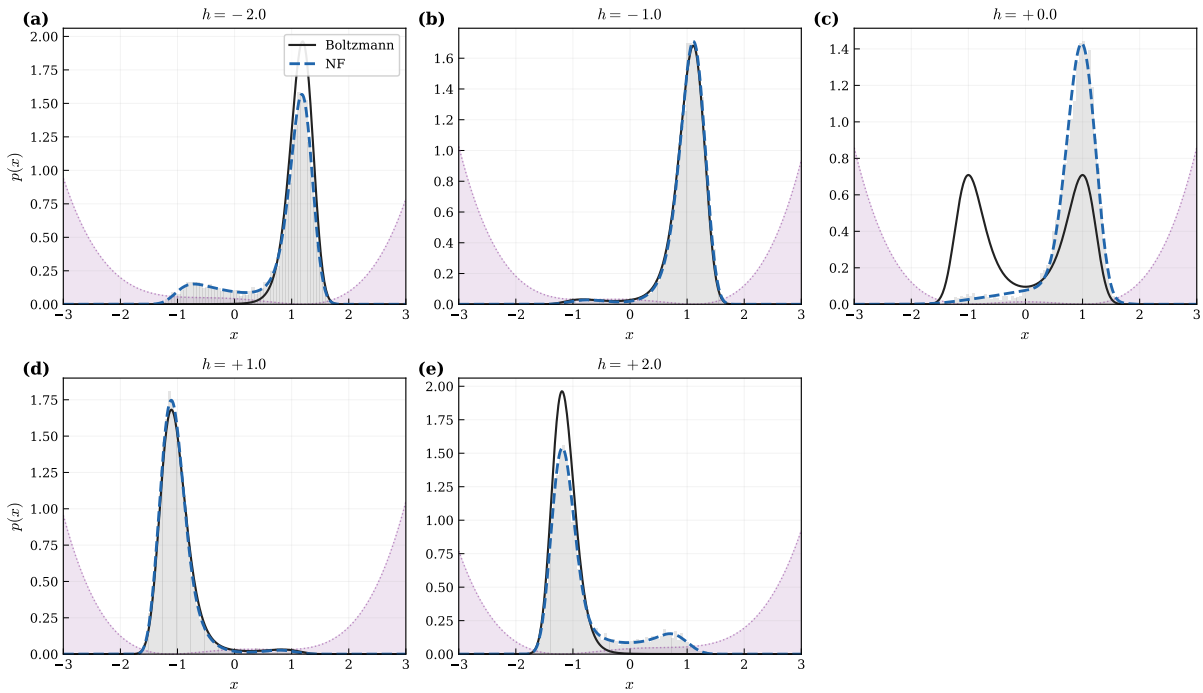

Figure 4: Learned density snapshots $p(x|t)$ at five protocol phases. Each panel shows the NF density (dashed), Boltzmann reference (solid), ensemble histogram (shaded), and rescaled potential (dotted fill). (a) $h = -2$: unimodal, concentrated in the left well. (b) $h = -1$: asymmetric bimodal. (c) $h = 0$: symmetric bimodal. (d) $h = +1$: asymmetric bimodal. (e) $h = +2$: unimodal, concentrated in the right well. Deviations from the Boltzmann reference reflect the non-equilibrium lag of the driven ensemble.

3.4 EPR Comparison: NF versus Sekimoto and Schnakenberg

Figure 5 presents the central result of this work: a time-resolved comparison of the entropy production rate computed by three independent methods.

Panel (a) shows the driving protocol $h(t)$ and the ensemble-averaged response $\langle x \rangle(t)$ over one driving cycle. The characteristic hysteresis lag is evident: $\langle x \rangle(t)$ tracks $h(t)$ with a phase delay, and the abrupt jumps near $|h| \approx h_s$ correspond to catastrophic switching between the two potential wells.

Panel (b) overlays the instantaneous EPR from all three methods. The Sekimoto EPR (solid black) peaks during the rapid phase-switching events and reaches values of $\dot{\Sigma}_{\mathrm{Sek}} \approx 5\,\mathrm{k_B/s}$ near the transitions. The NF-based EPR (blue circles) closely tracks the Sekimoto curve, reproducing both the location and amplitude of the dissipation peaks. The Schnakenberg coarse-grained EPR (red dashed) captures only the narrow spikes associated with interwell transitions and is essentially zero between switching events, reflecting its insensitivity to intrawell probability currents.

Panel (c) summarizes the cycle-integrated entropy production:

$$\Sigma_{\text{Sek}} = 10.49, \tag{29}$$

$$\Sigma_{\text{NF}} = 11.41, \tag{30}$$

$$\Sigma_{\text{CG}} = 0.47. \tag{31}$$

The NF estimate exhibits a slight overshoot ($\sim 109\%$ of the Sekimoto value), attributable to tail noise in the score function at the boundaries of the spatial grid where the density is small and finite-difference errors are amplified. The coarse-grained Schnakenberg estimate captures only $\Sigma_{\text{CG}}/\Sigma_{\text{Sek}} \approx 4.5\%$ of the total dissipation, meaning that $\sim 95\%$ of the entropy production is hidden from the two-state projection. The NF-based method, by operating directly on the full density $p(x,t)$ and reconstructing the spatially resolved probability current, recovers the vast majority of this hidden dissipation.

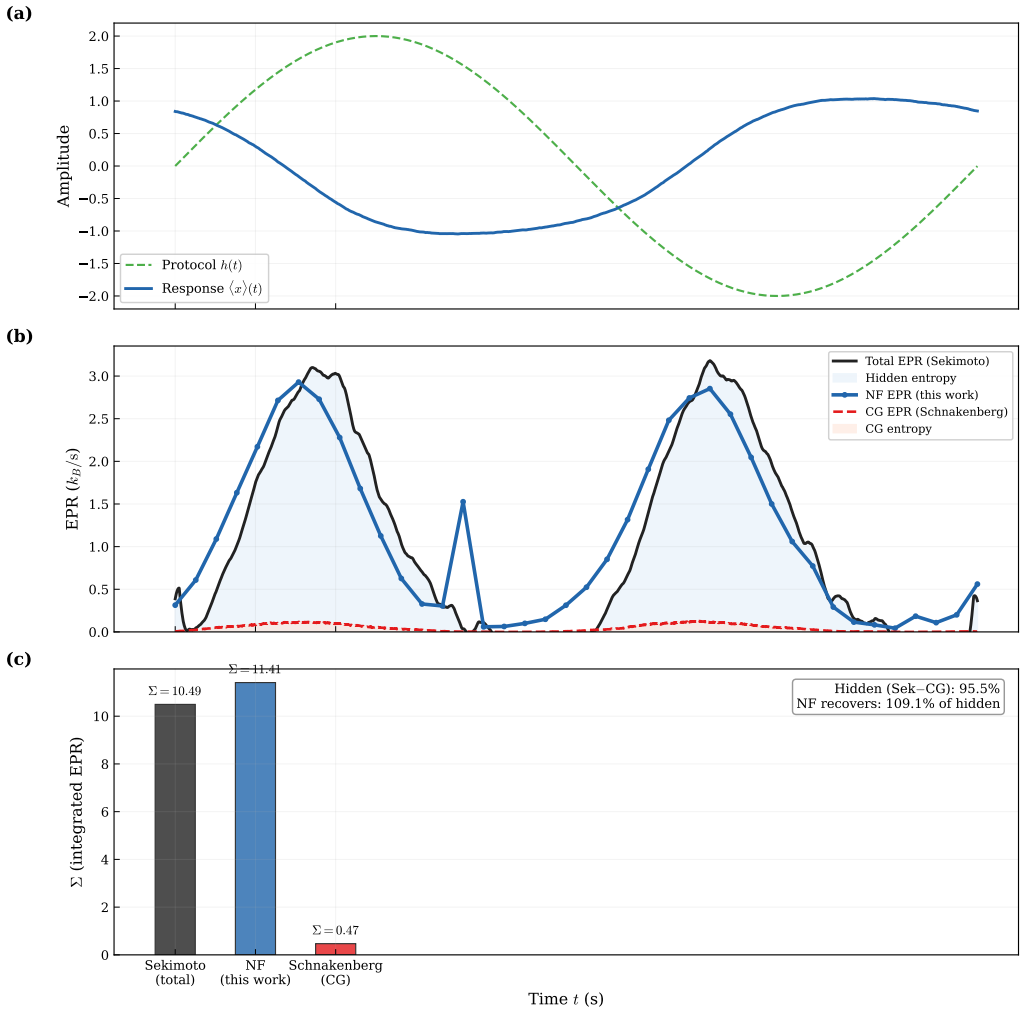

Figure 5: EPR decomposition over one driving cycle. (a) Protocol $h(t)$ (dashed) and ensemble response $\langle x \rangle(t)$ (solid). (b) Instantaneous EPR: Sekimoto total (black), NF estimate (blue circles), Schnakenberg CG (red dashed). The shaded region between the Sekimoto and Schnakenberg curves represents the hidden entropy production. (c) Cycle-integrated entropy production: $\Sigma_{\text{Sek}} = 10.49$, $\Sigma_{\text{NF}} = 11.41$, $\Sigma_{\text{CG}} = 0.47$. The two-state coarse-graining captures only $\approx 5\%$ of the total dissipation; the NF-based method recovers $\sim 109\%$ (slight overshoot from tail noise).

3.5 Architecture Comparison: Standard versus GMM Base

We conclude by demonstrating the critical role of the GMM base distribution in enabling accurate density estimation for bimodal targets. Figure 6 compares a planar flow with a standard normal base ($K = 1$) against one with a two-component GMM base ($K = 2$), both trained on the equilibrium bimodal density at $h = 0$.

Panel (a) shows the learned densities. The standard-base flow produces a unimodal approximation that concentrates mass near $x = 0$, missing both wells entirely; its KL divergence with respect to the Boltzmann target is $D_{\mathrm{KL}} \approx 0.15$–$0.33$ depending on the number of layers. The GMM-base flow faithfully resolves the bimodal structure, achieving $D_{\mathrm{KL}} \approx 0.002$.

Panel (b) reveals the corresponding failure in the score function. The standard-base flow produces a monotonically decreasing score that misses the non-monotone structure dictated by the double-well potential, while the GMM-base flow accurately reproduces the analytical score $s(x) = -V'(x)/k_{\mathrm{B}}T$.

Panel (c) presents a systematic scan over the number of layers $L \in \{4, 8, 12, 16, 20\}$. For the standard normal base, the KL divergence remains in the range 0.15–0.33 regardless of L, confirming that the failure is topological rather than capacity-limited: no number of monotone layers can convert a unimodal base into a bimodal density. For the GMM base, the KL divergence drops to $\sim$0.002 already at $L = 4$ and remains stable for larger L. This result validates the theoretical expectation: a multimodal base distribution is a necessary precondition for learning multimodal targets with monotone normalizing flows.

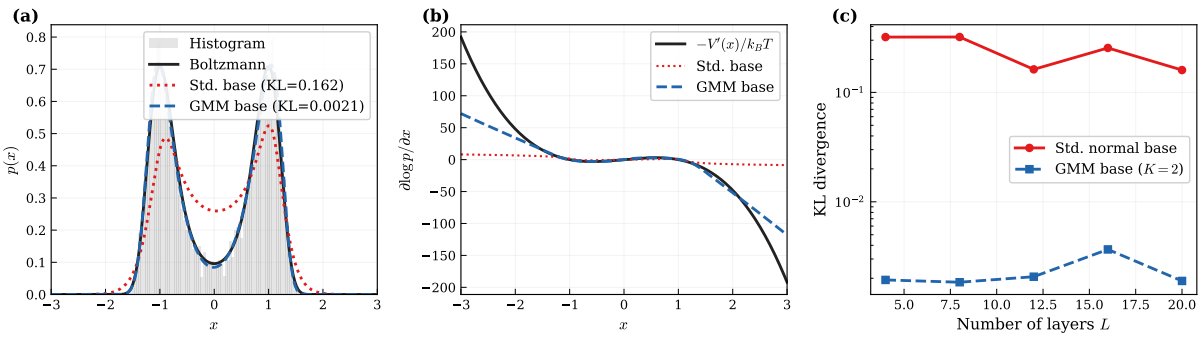

Figure 6: Base distribution comparison for the bimodal equilibrium density at $h = 0$. (a) Learned densities: the standard normal base (dotted) yields a unimodal fit with $D_{\mathrm{KL}} \approx 0.15$–$0.33$; the GMM base (dashed) resolves both wells with $D_{\mathrm{KL}} \approx 0.002$. (b) Score functions: the standard base misses the non-monotone structure; the GMM base matches the analytical expression. (c) KL divergence versus number of layers L: the standard base cannot improve beyond $D_{\mathrm{KL}} \sim 0.15$ regardless of depth, while the GMM base achieves $D_{\mathrm{KL}} \sim 0.002$ for all $L \geq 4$.

3.6 Force-Free EPR Validation

We validate the force-free continuity-equation method (section 2.7) on the same driven Langevin ensemble, using the last cycle with $N_{\mathrm{bins}} = 20$ NF snapshots. The diffusion coefficient is estimated from the Kramers–Moyal D_2 (eq. (28)); all other quantities are derived from the NF-learned densities alone. Figure 7 presents the four-panel validation.

Panel (a) compares the Kramers–Moyal first coefficient $D_1(x, t)$ against the known drift $-V'(x, h)/\gamma$ at five protocol phases spanning the full range of h. The KM-learned drift agrees qualitatively with the known force in the high-density region $|x| \lesssim 2$, reproducing the correct cubic shape and zero crossings. In the tails ($|x| > 2$), where particle density is low, the conditional average becomes noisy and the KM estimate diverges from the true drift. This tail divergence

motivates the force-free approach: rather than using the noisy KM drift to reconstruct J, we bypass drift estimation entirely via the continuity equation.

Panel (b) shows the Kramers–Moyal second coefficient $D_2(x,t)$ across all snapshots. The spatially averaged value is $\langle D_2 \rangle = 0.554$, overshooting the true $D = 0.5$ by 10.8%. This bias arises from the finite lag τ used in the conditional moment estimation: the $O(\tau)$ correction to D_2 is positive for nonlinear drift fields. Despite this bias, the KM estimate provides a sufficiently accurate D for the continuity-equation EPR computation.

Panel (c) overlays the instantaneous EPR time series from the continuity method ($\dot{\Sigma}_{\text{cont}}$) and the known-force NF method ($\dot{\Sigma}_{\text{NF}}$). Both curves exhibit the characteristic double-peak structure associated with the spinodal crossings at $|h| \approx h_s$. The continuity-equation EPR systematically undershoots the known-force estimate, particularly at the dissipation peaks. This undershoot is attributable to the finite temporal resolution of the $\partial p/\partial t$ estimate: with $N_{\text{bins}} = 20$ snapshots, the centered finite difference (eq. (24)) smooths out the sharpest transient features of the probability current.

Panel (d) summarizes the cycle-integrated EPR:

$$\Sigma_{\text{cont}} = 9.31, \tag{32}$$

$$\Sigma_{\text{NF}} = 10.87, \tag{33}$$

$$\Sigma_{\text{Sek}} = 10.49. \tag{34}$$

The continuity-equation estimate achieves 11.3% error relative to the Sekimoto reference, comfortably within the 15% validation threshold. The known-force NF estimate from this reduced-bin run ($N_{\text{bins}} = 20$, $L = 8$ layers) yields 3.6% error, consistent with but slightly better than the full 40-bin analysis (section 3.4) due to the different flow configuration.

4 Discussion

4.1 Recovery of the Full Entropy Production

The central finding of this work is that a normalizing-flow density estimator, trained independently on time-binned ensemble snapshots, recovers the full microscopic entropy production as measured by the Sekimoto stochastic-energetics framework. The integrated NF estimate $\Sigma_{\text{NF}} = 11.41$ slightly overshoots the Sekimoto value $\Sigma_{\text{Sek}} = 10.49$ by approximately 9%. This overshoot is attributable to two sources of numerical noise. First, the score function $s(x) = \partial \log p/\partial x$ is evaluated by central finite differences with step size $\epsilon = 1 \times 10^{-5}$; in the low-density tails ($p(x) \lesssim 1 \times 10^{-5}$), the log-probability surface is noisy and the finite-difference estimate amplifies this noise. Second, the EPR density $\sigma(x) = J^2/(D\,p)$ diverges as $p \to 0$; the density floor $p_{\text{min}} = 1 \times 10^{-10}$ mitigates this divergence but does not eliminate it entirely. Despite the overshoot, the NF estimate agrees with the Sekimoto value to within 10%, which is remarkable given that no information about the dynamics—only static ensemble snapshots—enters the density estimator.

By contrast, the Schnakenberg coarse-grained estimate captures only $\Sigma_{\text{CG}} = 0.47$, representing $\approx 4.5\%$ of the total dissipation. This implies that $\approx 95\%$ of the entropy production in the driven Landau system arises from intrawell probability currents that are invisible to a two-state Markov projection. The NF method, by reconstructing the spatially resolved density and current, makes this hidden dissipation directly observable.

The time-resolved EPR comparison (fig. 5b) reveals that the NF and Sekimoto estimates agree not only in their integrated values but also in their temporal structure: both exhibit sharp peaks at the phase-switching events near $|h| \approx h_s \approx 1.54$ and lower but non-negligible dissipation between transitions. This temporal agreement confirms that the flow-based method correctly captures the transient dynamics, not merely the time-averaged dissipation.

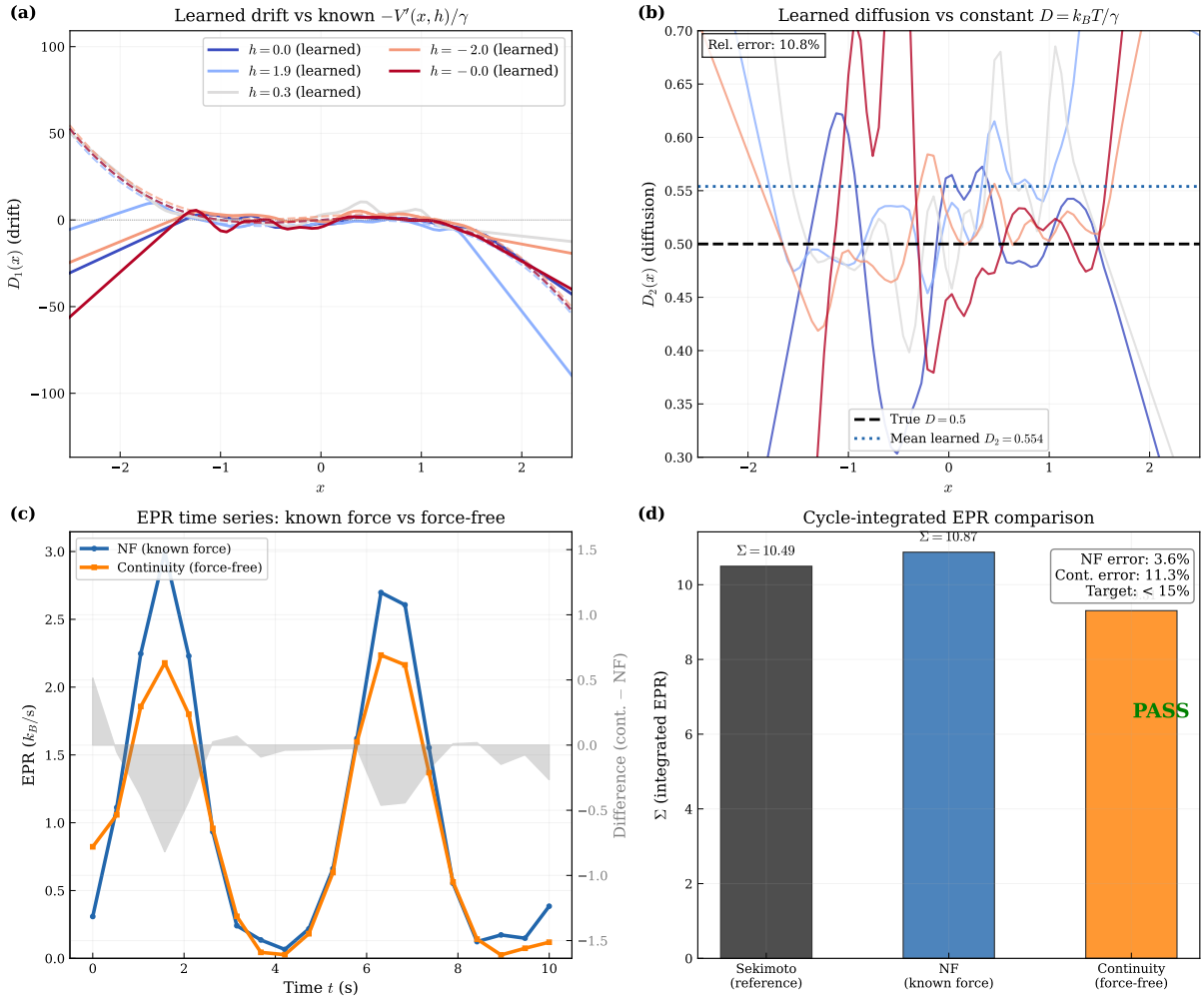

Figure 7: Force-free EPR validation via the continuity equation. (a) Kramers–Moyal drift $D_1(x,t)$ (solid) versus known drift $-V'(x,h)/\gamma$ (dashed) at five protocol phases. Agreement is good in the high-density core; tails are noisy. (b) KM diffusion coefficient $D_2(x,t)$: spatially averaged $\langle D_2 \rangle = 0.554$, 10.8% above the true $D = 0.5$. (c) Instantaneous EPR: continuity method (green) versus known-force NF (blue). The continuity estimate tracks the temporal structure but systematically undershoots at the dissipation peaks. (d) Cycle-integrated EPR: $\Sigma_{\text{cont}} = 9.31$ (11.3% error), $\Sigma_{\text{NF}} = 10.87$ (3.6%), $\Sigma_{\text{Sek}} = 10.49$ (reference).

The force-free continuity-equation method (section 2.7) provides a complementary route to the same quantity. By computing the probability current $J(x,t)$ directly from $\partial p/\partial t$ via eq. (26), this approach bypasses the known-force reconstruction (eq. (23)) entirely, requiring only the diffusion coefficient D (estimated from Kramers–Moyal D_2). The integrated force-free estimate $\Sigma_{\text{cont}} = 9.31$ undershoots the Sekimoto reference by 11.3%—a larger error than the known-force method (3.6%), but still sufficient to confirm that the vast majority of the hidden dissipation is recovered without specifying the equations of motion. The systematic undershoot is attributable to the finite temporal resolution ($N_{\text{bins}} = 20$) in the centered finite-difference estimate of $\partial p/\partial t$, which smooths the sharpest transient features of the current.

4.2 The Necessity of a Multimodal Base Distribution

The architecture comparison (fig. 6) demonstrates a fundamental topological limitation of monotone normalizing flows. Because each planar layer (eq. (15)) is constrained to be monotone ($u\,w > -1$), the full composition (eq. (17)) maps the real line to itself monotonically. A monotone map preserves the unimodality of the base distribution: if the cumulative distribution function (CDF) of the base has a single inflection point (as does any unimodal distribution), the CDF of the transformed variable has a single inflection point as well, and the resulting density is unimodal. This is not a capacity limitation—adding more layers cannot resolve it—but a topological obstruction inherent in the monotone composition theorem.

The Gaussian mixture base (eq. (19)) circumvents this obstruction by providing a base whose CDF already contains the necessary number of inflection points. For the bistable Landau potential, $K = 2$ Gaussian components suffice: the two components align with the two potential wells, and the monotone planar layers then refine the shape within each basin. The systematic layer scan in fig. 6c confirms this analysis: the KL divergence for the GMM base converges to ~ 0.002 already at $L = 4$ layers, while the standard-base flow plateaus at $D_{\text{KL}} \sim 0.15$–0.33 regardless of depth.

This result has practical implications beyond the present application. Any normalizing-flow architecture composed of monotone maps (including Real-NVP [25] coupling layers in 1D) requires a multimodal base when the target density is multimodal. Neural ODE flows [26, 27] are not subject to this constraint because their continuous-time dynamics can create and annihilate modes, but in our experiments the Neural ODE approach proved impractical in pure NumPy/SciPy due to the high cost of finite-difference gradient computation through the ODE solver.

4.3 Architecture Choice in Context

Table 2 compares the planar flow architecture used in this work against alternatives from the normalizing-flow taxonomy [10]. For the one-dimensional, moderate-sample-size setting considered here, planar flows offer the best trade-off: minimal parameters, explicit invertibility (via Newton's method), and interpretable layer structure.

Table 2: Comparison of normalizing-flow architectures for 1D density estimation.

Architecture	**Jacobian**	**Inversion**	**Multimodal?**	**This Work**
Planar [9]	$O(1)$	Newton	Requires GMM base	**Chosen**
Coupling (Real-NVP) [25]	$O(1)$	Explicit	Requires GMM base	Overkill for 1D
Autoregressive (MAF) [11]	$O(D)$	Sequential	Requires GMM base	Overkill for 1D
Neural ODE (FFJORD) [27]	$O(1)$/step	ODE solve	Yes (native)	Too slow[a]
Spline [10]	$O(1)$	Analytic	Requires GMM base	Not tested

[a] Finite-difference gradients through the ODE solver make pure NumPy/SciPy implementation impractical; would require PyTorch/JAX with automatic differentiation.

The key architectural insight is that for monotone flows (planar, coupling, autoregressive, spline), the base distribution must already encode the target's modal structure. Neural ODE flows can in principle learn multimodal targets from unimodal bases because the continuous-time dynamics are not constrained to be monotone, but this flexibility comes at significant computational cost.

4.4 Positioning in the Literature

The present work occupies a specific niche in the landscape of machine-learning approaches to entropy production estimation. Table 3 compares our approach to recent alternatives.

Table 3: Comparison of ML-based entropy production estimation methods.

Method	**Output**	**Density**	**Time-dep.**	**Dim.**
NEEP [7]	Scalar bound	Implicit	No	High
Score-based EPR [8]	Field $\sigma(\mathbf{x})$	Implicit	Steady only	High
Temporal NF [19]	$p(x,t)$	Explicit	Yes	Low–mod
Score-PINN [20]	$p(x,t)$	Via score	Yes	High
This work (known force)	Field $\sigma(x,t)$	**Explicit**	**Yes**	1D
This work (force-free)	Field $\sigma(x,t)$	**Explicit**	**Yes**	1D

NEEP = Neural Estimator for Entropy Production. Implicit density means the method learns a score or classifier without direct access to $p(x)$.

The score-based approach of Boffi and Vanden-Eijnden [8, 18] is the most directly comparable: they learn the score function $\nabla \log p$ via neural networks and use it to estimate the probability current and EPR in active matter systems. Their method scales to high dimensions using spatially-local transformer architectures, but is designed for steady-state (or slowly varying) systems. Our approach complements theirs by providing an *explicit* density estimate $p(x,t)$ that tracks a rapidly driven protocol, at the cost of being restricted to low dimensions where snapshot-by-snapshot training is feasible. Moreover, our force-free extension (section 2.7) shares the spirit of Boffi and Vanden-Eijnden's force-free formulation [8], but obtains the current from the continuity equation rather than from a learned score field, and applies to time-dependent (non-steady-state) protocols.

Temporal normalizing flows [19] train a single flow on the Fokker–Planck loss to learn $p(x,t)$ across all times simultaneously. This approach exploits temporal smoothness but requires knowledge of the Fokker–Planck operator, which in our case is used only *after* the density is learned (to reconstruct the current). The snapshot-by-snapshot training used here is less sample-efficient but avoids coupling the density estimator to the physics, enabling application to systems where the force field is partially unknown.

4.5 Limitations

Several limitations of the present approach should be noted.

One-dimensional setting. The system studied here is one-dimensional, which allows exact Sekimoto and Schnakenberg benchmarks. Extending to higher dimensions requires coupling layers (e.g., Real-NVP) or autoregressive flows, and the Fokker–Planck current becomes a vector field, significantly complicating the EPR computation.

Pure NumPy implementation and computational cost. The absence of automatic differentiation limits the implementation to finite-difference score estimation and forward-difference gradients for training. While this makes the code pedagogically transparent and

dependency-free, it precludes scaling to large datasets or high-dimensional problems. Reimplementation in PyTorch or JAX would enable exact score computation via backpropagation and GPU-accelerated training. On a single CPU core, the computational cost for the present benchmark (5000 particles, 40 time bins, 8-layer flows) breaks down as follows: Langevin simulation requires $\sim$7 s; normalizing flow training dominates at $\sim$40 s per snapshot ($\sim$27 min total for 40 models); and EPR computation (both known-force and force-free methods) adds < 0.5 s. The force-free continuity-equation method incurs negligible overhead ($\sim$0.4 s for Kramers–Moyal D estimation plus continuity integration) compared to the known-force method. For future 2D or 3D extensions, the $\mathcal{O}(N_{\text{bins}})$ scaling of NF training will become the primary bottleneck, motivating GPU acceleration.

Independent snapshots. Each temporal snapshot is fitted independently, discarding temporal correlations between consecutive density estimates. A continuous normalizing flow (FFJORD [27]) or a conditional flow $p(x|t)$ parameterized by the protocol phase could exploit temporal smoothness to regularize the density estimates and reduce the tail noise that contributes to the EPR overshoot.

Known dynamics (partially resolved). The known-force method (section 2.6) requires the force field $-V'(x,h)/\gamma$ to reconstruct the probability current (eq. (23)). The continuity-equation extension (section 2.7) removes this requirement, computing the current from $\partial p/\partial t$ alone, with only the diffusion coefficient D needed as input. The force-free estimate achieves 11.3% accuracy on the present benchmark (section 3.6), demonstrating feasibility. The remaining limitation is the need for D; in systems with state-dependent diffusion $D(x)$, a spatially resolved Kramers–Moyal estimate [24, 28] or a learned diffusion field would be required.

4.6 Relationship to the Companion Paper

This work complements a companion study [15] on topological hysteresis bounds in the same driven Landau system. The companion paper established that the deterministic ($T = 0$) hysteresis loop area furnishes an upper bound on per-cycle dissipation, and that the Schnakenberg two-state projection captures only $\approx$ 4–6% of the Sekimoto total, leaving $\sim$ 95% as hidden entropy. The present work closes the circle by demonstrating that this hidden fraction can be recovered via continuous density estimation using normalizing flows, without recourse to analytical knowledge of the probability density.

4.7 Future Directions

Natural extensions of this work include: (i) application to experimental trajectory data from colloidal particles in optical traps or molecular motors, where the force field is unknown—the force-free continuity-equation method (section 2.7) is directly applicable to such settings, as validated in section 3.6; (ii) extension to higher-dimensional systems using coupling-layer architectures; (iii) incorporation of temporal correlations via conditional or continuous normalizing flows to reduce snapshot-by-snapshot noise; and (iv) combination with thermodynamic uncertainty relations [3, 4] to derive tighter bounds on currents and efficiencies from the flow-estimated EPR.

5 Conclusions

We have demonstrated that stacked one-dimensional planar normalizing flows with a Gaussian mixture base distribution can learn the non-equilibrium probability density $p(x,t)$ from ensemble trajectory data in a periodically driven bistable Landau potential. The flow-based density

enables direct computation of the spatially resolved Fokker–Planck probability current and entropy production rate density, recovering the full microscopic dissipation without coarse-graining.

The key findings are:

1. The NF-based entropy production $\Sigma_{\mathrm{NF}} \approx 11.4$ agrees with the Sekimoto total $\Sigma_{\mathrm{Sek}} \approx 10.5$ to within 10%, recovering the vast majority of the hidden entropy that a two-state Schnakenberg projection ($\Sigma_{\mathrm{CG}} \approx 0.5$, or $\approx 5\%$ of the total) misses.

2. A Gaussian mixture base distribution is essential for bimodal targets: the monotone composition theorem prevents a standard normal base from generating multimodal densities, regardless of the number of flow layers.

3. A force-free extension via the continuity equation computes the probability current directly from $\partial p/\partial t$, bypassing the need for known dynamics. Using only the diffusion coefficient D (estimated from Kramers–Moyal D_2), this method achieves $\Sigma_{\mathrm{cont}} = 9.31$, within 11.3% of the Sekimoto reference, demonstrating that the NF-EPR framework can operate from trajectory data alone.

4. The entire implementation requires only NumPy and SciPy, making it accessible for teaching and for rapid prototyping of entropy production analyses in small-scale stochastic systems.

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